

Humanoid Robot Market Expected to Reach \$5 Trillion by 2050 | Morgan Stanley

Source HTML: 机器人产品资料库_2026-06-21/01_Excel链接资料/市场与行业报告/007_7_-_投资机构____人形机器人_-_Humanoids__A_\$5_Trillion_Market.html

- Investor Relations
- Media Relations
- Alumni
- Client Login Client Login Morgan Stanley Online Morgan Stanley at Work Research Portal Matrix Morgan Stanley Online Morgan Stanley at Work Research Portal Matrix
- Morgan Stanley Online
- Morgan Stanley at Work
- Research Portal
- Matrix

Morgan Stanley Online

Morgan Stanley at Work

Research Portal

Matrix

- Our Insights Our Insights Personal Finance Market Trends Technology & Disruption Sustainability Diversity & Inclusion Morgan Stanley Institute STAY UP TO DATE Sign Up For the Five Ideas Newsletter Listen To Our Podcasts At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Personal Finance Learn from our industry leaders about how to manage your wealth and help meet your personal financial goals. At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Market Trends From volatility and geopolitics to economic trends and investment outlooks, stay informed on the key developments shaping today's markets. At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Technology & Disruption Whether it ' s hardware, software or age-old businesses, everything today is ripe for disruption. Stay abreast of the latest trends and developments. At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Sustainability Our insightful research, advisory and investing capabilities give us unique and broad perspective on sustainability topics. At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Diversity & Inclusion Multicultural and women entrepreneurs are the cutting-edge leaders of businesses that power markets. Hear their stories and learn about how they are redefining the terms of success. At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Morgan Stanley Institute Signal that moves capital. The Morgan Stanley Institute delivers integrated insights on the defining questions for financial decision makers. Featured 2026 Midyear Outlooks Insights Hard Lessons Insights Energy: Transitioning to Resilience Morgan Stanley Institute
- Personal Finance
- Market Trends
- Technology & Disruption
- Sustainability
- Diversity & Inclusion
- Morgan Stanley Institute
- Sign Up For the Five Ideas Newsletter
- Listen To Our Podcasts

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Personal Finance Learn from our industry leaders about how to manage your wealth and help meet your personal financial goals.

At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues.

Personal Finance

Learn from our industry leaders about how to manage your wealth and help meet your personal financial goals.

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Market Trends From volatility and geopolitics to economic trends and investment outlooks, stay informed on the key developments shaping today's markets.

Market Trends

From volatility and geopolitics to economic trends and investment outlooks, stay informed on the key developments shaping today's markets.

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Technology & Disruption Whether it ' s hardware, software or age-old businesses, everything today is ripe for disruption. Stay abreast of the latest trends and developments.

Technology & Disruption

Whether it ' s hardware, software or age-old businesses, everything today is ripe for disruption. Stay abreast of the latest trends and developments.

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Sustainability Our insightful research, advisory and investing capabilities give us unique and broad perspective on sustainability topics.

Sustainability

Our insightful research, advisory and investing capabilities give us unique and broad perspective on sustainability topics.

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Diversity & Inclusion Multicultural and women entrepreneurs are the cutting-edge leaders of businesses that power markets. Hear their stories and learn about how they are redefining the terms of success.

Diversity & Inclusion

Multicultural and women entrepreneurs are the cutting-edge leaders of businesses that power markets. Hear their stories and learn about how they are redefining the terms of success.

- At Morgan Stanley, we lead with exceptional ideas. Across all our businesses, we offer keen insight on today's most critical issues. Morgan Stanley Institute Signal that moves capital. The Morgan Stanley Institute delivers integrated insights on the defining questions for financial decision makers.

Morgan Stanley Institute

Signal that moves capital. The Morgan Stanley Institute delivers integrated insights on the defining questions for financial decision makers.

- What We Do What We Do Wealth Management Investment Banking & Capital Markets Sales & Trading Research Investment Management Morgan Stanley at Work Sustainable Investing Inclusive & Sustainable Ventures Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Wealth Management We help people, businesses and institutions build, preserve and manage wealth so they can pursue their financial goals. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Investment Banking & Capital Markets We have global expertise in market analysis and in advisory and capital-raising services for corporations, institutions and governments. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Sales & Trading Global institutions, leading hedge funds and industry innovators turn to Morgan Stanley for sales, trading and market-making services. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Research We offer timely, integrated analysis of companies, sectors, markets and economies, helping clients with their most critical decisions. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Investment Management We deliver active investment strategies

across public and private markets and custom solutions to institutional and individual investors. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Morgan Stanley at Work We provide comprehensive workplace financial solutions for organizations and their employees, combining personalized advice with modern technology. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Sustainable Investing We offer scalable investment products, foster innovative solutions and provide actionable insights across sustainability issues. Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Inclusive & Sustainable Ventures From our startup lab to our cutting-edge research, we broaden access to capital for entrepreneurs and spotlight their success. Featured Get Matched with a Financial Advisor Wealth Management Explore What We Offer E*TRADE from Morgan Stanley Watch Our Client Success Stories The Morgan Stanley Blueprint

- Wealth Management
- Investment Banking & Capital Markets
- Sales & Trading
- Research
- Investment Management
- Sustainable Investing
- Inclusive & Sustainable Ventures
- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Wealth Management We help people, businesses and institutions build, preserve and manage wealth so they can pursue their financial goals.

Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals.

Wealth Management

We help people, businesses and institutions build, preserve and manage wealth so they can pursue their financial goals.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Investment Banking & Capital Markets We have global expertise in market analysis and in advisory and capital-raising services for corporations, institutions and governments.

Investment Banking & Capital Markets

We have global expertise in market analysis and in advisory and capital-raising services for corporations, institutions and governments.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Sales & Trading Global institutions, leading hedge funds and industry innovators turn to Morgan Stanley for sales, trading and market-making services.

Sales & Trading

Global institutions, leading hedge funds and industry innovators turn to Morgan Stanley for sales, trading and market-making services.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Research We offer timely, integrated analysis of companies, sectors, markets and economies, helping clients with their most critical decisions.

Research

We offer timely, integrated analysis of companies, sectors, markets and economies, helping clients with their most critical decisions.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Investment Management We deliver active investment strategies across public and private markets and custom solutions to institutional and individual investors.

Investment Management

We deliver active investment strategies across public and private markets and custom solutions to institutional and individual investors.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Morgan Stanley at Work We provide comprehensive workplace financial solutions for organizations and their employees, combining personalized advice with modern technology.

We provide comprehensive workplace financial solutions for organizations and their employees, combining personalized advice with modern technology.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Sustainable Investing We offer scalable investment products, foster innovative solutions and provide actionable insights across sustainability issues.

Sustainable Investing

We offer scalable investment products, foster innovative solutions and provide actionable insights across sustainability issues.

- Morgan Stanley helps people, institutions and governments raise, manage and distribute the capital they need to achieve their goals. Inclusive & Sustainable Ventures From our startup lab to our cutting-edge research, we broaden access to capital for entrepreneurs and spotlight their success.

Inclusive & Sustainable Ventures

From our startup lab to our cutting-edge research, we broaden access to capital for entrepreneurs and spotlight their success.

- About Us About Us Core Values Leadership History Locations Diversity Sustainability Giving Back Technology Sponsorships Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Core Values Everything we do at Morgan Stanley is guided by our five core values: Do the right thing, put clients first, lead with exceptional ideas, commit to diversity and inclusion, and give back. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Leadership Morgan Stanley leadership is dedicated to conducting first-class business in a first-class way. Our board of directors and senior executives hold the belief that capital can and should benefit all of society. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. History From our origins as a small Wall Street partnership to becoming a global firm of more than 80,000 employees today, Morgan Stanley has been committed to clients and communities for 90 years. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Locations The global presence that Morgan Stanley maintains is key to our clients' success, giving us keen insight across regions and markets, and allowing us to make a difference around the world. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Diversity Morgan Stanley is differentiated by the caliber of our diverse team. Our culture of access and inclusion has built our legacy and shapes our future, helping to strengthen our business and bring value to clients. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Sustainability Our firm's commitment to sustainability informs our operations, governance, risk management, diversity efforts, philanthropy and research. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Giving Back At Morgan Stanley, giving back is a core value—a central part of our culture globally. We live that commitment through long-lasting partnerships, community-based delivery and engaging our best asset—Morgan Stanley employees. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Technology As a global financial services firm, Morgan Stanley is committed to technological innovation. We rely on our technologists around the world to create leading-edge, secure platforms for all our businesses. Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Sponsorships At Morgan Stanley, we believe creating a more equitable society begins with investing in access, knowledge and resources to foster potential for all. We are committed to supporting the next generation of leaders and ensuring that they reflect the diversity of the world they inherit.

- Core Values
- Leadership
- History
- Locations
- Diversity
- Giving Back

- Technology

- Sponsorships

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Core Values Everything we do at Morgan Stanley is guided by our five core values: Do the right thing, put clients first, lead with exceptional ideas, commit to diversity and inclusion, and give back.

Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values.

Core Values

Everything we do at Morgan Stanley is guided by our five core values: Do the right thing, put clients first, lead with exceptional ideas, commit to diversity and inclusion, and give back.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Leadership Morgan Stanley leadership is dedicated to conducting first-class business in a first-class way. Our board of directors and senior executives hold the belief that capital can and should benefit all of society.

Leadership

Morgan Stanley leadership is dedicated to conducting first-class business in a first-class way. Our board of directors and senior executives hold the belief that capital can and should benefit all of society.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. History From our origins as a small Wall Street partnership to becoming a global firm of more than 80,000 employees today, Morgan Stanley has been committed to clients and communities for 90 years.

History

From our origins as a small Wall Street partnership to becoming a global firm of more than 80,000 employees today, Morgan Stanley has been committed to clients and communities for 90 years.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Locations The global presence that Morgan Stanley maintains is key to our clients' success, giving us keen insight across regions and markets, and allowing us to make a difference around the world.

Locations

The global presence that Morgan Stanley maintains is key to our clients' success, giving us keen insight across regions and markets, and allowing us to make a difference around the world.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Diversity Morgan Stanley is differentiated by the caliber of our diverse team. Our culture of access and inclusion has built our legacy and shapes our future, helping to strengthen our business and bring value to clients.

Diversity

Morgan Stanley is differentiated by the caliber of our diverse team. Our culture of access and inclusion has built our legacy and shapes our future, helping to strengthen our business and bring value to clients.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Sustainability Our firm's commitment to sustainability informs our operations, governance, risk management, diversity efforts, philanthropy and research.

Our firm's commitment to sustainability informs our operations, governance, risk management, diversity efforts, philanthropy and research.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Giving Back At Morgan Stanley, giving back is a core value—a central part of our culture globally. We live that commitment through long-lasting partnerships, community-based delivery and engaging our best asset—Morgan Stanley employees.

Giving Back

At Morgan Stanley, giving back is a core value—a central part of our culture globally. We live that commitment through long-lasting partnerships, community-based delivery and engaging our best asset—Morgan Stanley employees.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Technology As a global financial services firm, Morgan Stanley is committed to technological innovation. We rely on our technologists around the world to create leading-edge, secure platforms for all our businesses.

Technology

As a global financial services firm, Morgan Stanley is committed to technological innovation. We rely on our technologists around the world to create leading-edge, secure platforms for all our businesses.

- Since our founding in 1935, Morgan Stanley has consistently delivered first-class business in a first-class way. Underpinning all that we do are five core values. Sponsorships At Morgan Stanley, we believe creating a more equitable society begins with investing in access, knowledge and resources to foster potential for all. We are committed to supporting the next generation of leaders and ensuring that they reflect the diversity of the world they inherit.

Sponsorships

At Morgan Stanley, we believe creating a more equitable society begins with investing in access, knowledge and resources to foster potential for all. We are committed to supporting the next generation of leaders and ensuring that they reflect the diversity of the world they inherit.

- Careers Careers Students & Graduates Opportunities Experienced Professionals Opportunities Technology Professionals Experienced Financial Advisors We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Students & Graduates A career at Morgan Stanley means belonging to an ideas-driven culture that embraces new perspectives to solve complex problems. See how you can make meaningful contributions as a student or recent graduate at Morgan Stanley. We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Experienced Professionals At Morgan Stanley, you ' ll find trusted colleagues, committed mentors and a culture that values diverse perspectives, individual intellect and cross-collaboration. See how you can continue your career journey at Morgan Stanley. We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Experienced Financial Advisors At Morgan Stanley, our premier brand, robust resources and market leadership can offer you a new opportunity to grow your practice and continue to fulfill on your commitment to deliver tailored wealth management advice that helps your clients reach their financial goals. Featured Explore Our Culture Careers Yes, You Can Be a Tech Innovator at Morgan Stanley Technology
- Students & Graduates Opportunities
- Opportunities
- Experienced Professionals Opportunities Technology Professionals
- Technology Professionals
- Experienced Financial Advisors
- We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Students & Graduates A career at Morgan Stanley means belonging to an ideas-driven culture that embraces new perspectives to solve complex problems. See how you can make meaningful contributions as a student or recent graduate at Morgan Stanley.

We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you.

Students & Graduates

A career at Morgan Stanley means belonging to an ideas-driven culture that embraces new perspectives to solve complex problems. See how you can make meaningful contributions as a student or recent graduate at Morgan Stanley.

- We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Experienced Professionals At Morgan Stanley, you ' ll find trusted colleagues, committed mentors and a culture that values diverse perspectives, individual intellect and cross-collaboration. See how you can continue your career journey at Morgan Stanley.

Experienced Professionals

At Morgan Stanley, you ' ll find trusted colleagues, committed mentors and a culture that values diverse perspectives, individual intellect and cross-collaboration. See how you can continue your career journey at Morgan Stanley.

- We believe our greatest asset is our people. We value our commitment to diverse perspectives and a culture of inclusion across the firm. Discover who we are and the right opportunity for you. Experienced Financial Advisors At Morgan Stanley, our premier brand, robust resources and market leadership can offer you a new opportunity to grow your practice and continue to fulfill on your commitment to deliver tailored wealth management advice that helps your clients reach their financial goals.

Experienced Financial Advisors

At Morgan Stanley, our premier brand, robust resources and market leadership can offer you a new opportunity to grow your practice and continue to fulfill on your commitment to deliver tailored wealth management advice that helps your clients reach their financial goals.

Humanoids: A \$5 Trillion Market

The number of robots that resemble and act like humans is likely to reach nearly 1 billion by 2050, with China currently leading in development.

Key Takeaways

- The humanoids market could surpass \$5 trillion by 2050, including sales from supply chains and networks for repair, maintenance and support.
- Adoption of humanoids is likely to accelerate in the late 2030s with improved technology as well as greater regulatory and societal support.
- Although humanoids are still under development, there could be more than 1 billion by 2050, with 90% used for industrial and commercial purposes.
- Currently, strong government support is helping China to lead the race to develop the humanoids market.

Are “ humanoids ” —human-like robots with arms, legs, hands and brains powered by artificial intelligence—on their way to work in factories, stores and your own kitchen? Not yet—but they are quickly evolving, and the market for humanoids could be twice the size of the auto industry in the coming decades.

Morgan Stanley Research estimates the humanoids market is likely to reach \$5 trillion by 2050, plus related supply chains as well as repair, maintenance and support. There could be more than 1 billion humanoids in use by 2050.

“ Adoption should be relatively slow until the mid-2030s, accelerating in the late 2030s and 2040s, ” says Adam Jonas, Morgan Stanley ’ s Head of Global Autos and Shared Mobility Research.

WATCH: AI Is About to Get Physical

Easier Commercial and Industrial Use

By 2050, about 90% of humanoids, or about 930 million units, will likely be used for repetitive, simple, and structured work—primarily industrial and commercial purposes. China is likely to have the highest number of humanoid robots in use by 2050, at 302.3 million, trailed by the U.S. at 77.7 million (up from the previous forecast of 63 million).

“ The forecast for household usage is much more conservative, with only 80 million humanoids in homes by 2050, ” Jonas says. “ We are not going to see a robot in every home overnight. ”

Creating a general-purpose humanoid that is capable of doing a vast array of useful tasks at home will require technological progress in both hardware and AI models, which should take about another decade. To get those humanoids into homes, prices need to decline significantly, in parallel with regulatory and societal acceptance of this use of humanoids.

“ Once we get to that stage, humanoid volume and penetration should pick up quickly, ” Jonas says.

Prices Are Likely to Fall

The complexity of humanoids, which require sophisticated robot software models and tight integration with hardware, makes them an expensive product. Morgan Stanley Research estimates that the cost of one humanoid was around \$200,000 in 2024 in high-income

countries.

As the technology advances and production volumes increase, prices are likely to fall to about \$150,000 by 2028 and \$50,000 by 2050. In lower-income countries, which may take more advantage of the cheaper Chinese supply chain, prices could fall to as low as \$15,000 by 2050.

At that point, the U.S. penetration rate could range from 3%, for households earning between \$50,000 and \$75,000 a year, to 33%, for those with annual income above \$200,000. About 10% of U.S. households overall could have a humanoid by 2050, totaling 15 million units in the country. Although China is likely to have the larger number of humanoids, only 3% of households are likely to own a humanoid, totaling about 4 million units.

“ We recognize that, hypothetically, the average household could have more than one unit, creating a fleet of humanoid butlers, ” Jonas says. “ However, in our forecasts, we assume one humanoid per household at this time. ”

China in the Lead

Investors may want to watch the fact that China is dominating the field of AI-enabled robotics, humanoids or otherwise, and the gap with the U.S. is widening.

“ It is becoming apparent that national support for ‘ embodied AI ’ may be far greater in China than in any other nation, driving continued innovation and capital formation, ” says Sheng Zhong, Morgan Stanley ’ s Head of Industrials Research. “ In our opinion, China's lead in AI-robotics may need to widen before rivals, including the U.S., pay closer attention. ”

Chinese supply-chain players are currently working on different solutions to improve the performance level of their components, via new design structures, new materials, refined manufacturing processes and AI algorithms to address the precision gap.

“ There are some leading U.S. players in humanoid design and development at this stage, but China could catch up when humanoids reach downstream application and mass production, riding on its strong self-sufficient supply chain, ” Zhong says.

Meanwhile, there are few U.S.-based alternatives for many humanoid components, such as screws, reducers, motors and batteries. Nearly every robot developer in the world still requires critical components sourced from China and other parts of Asia.

“ While it is too soon to declare a final champion in the race for agentic humanoid robot supremacy, the U.S. will need to make significant changes in manufacturing capability, education and national policies to remain competitive in this area, ” Jonas says.

For deeper insights and analysis, ask your Morgan Stanley Representative or Financial Advisor for the full report, “ A \$5 Trillion Global Market, ” (April 29, 2025).

Sign up to get Morgan Stanley Ideas delivered to your inbox.

Thank You for Subscribing!

Would you like to help us improve our coverage of topics that might interest you? Tell us about yourself.

- Disclosures
- Corporate Governance
- Newsroom
- Careers
- Contact Us
- Global Offices
- Equal Employment Opportunity
- Cybersecurity
- Terms of Use
- Privacy & Cookies
- Your Privacy Choices

© 2026 Morgan Stanley. All rights reserved.